Beam ID: C-1 | Units: kip, kip-ft, in | Analysis: shape

1. General Information

Beam ID	C-1
Design code	ACI 318-19
Run option	Factored PMM strength
Capacity method	Radial contour intersection at constant Pu
Concrete integration	Shape
Neutral-axis increment	10.000 deg
Slenderness	Not considered

2. Material Properties

2.1 Concrete

Property	Value
Concrete strength, f'c	4.000 ksi
Maximum compression strain, eps_cu	0.003 in/in
Equivalent block stress	3.400 ksi
Equivalent block beta1	0.850

2.2 Reinforcing steel

Property	Value
Yield strength, fy	60.000 ksi
Elastic modulus, Es	29,000 ksi
Yield strain, eps_y	0.0020690 in/in
Stress model	Elastic-perfectly plastic

3. Section

Property	Value
Section type	Circular
Diameter	24.000 in
Gross area, Ag	452.066 in2
Ix	16286.016 in4
Iy	16286.016 in4
Reference centroid	x = 0.000 in, y = 0.000 in

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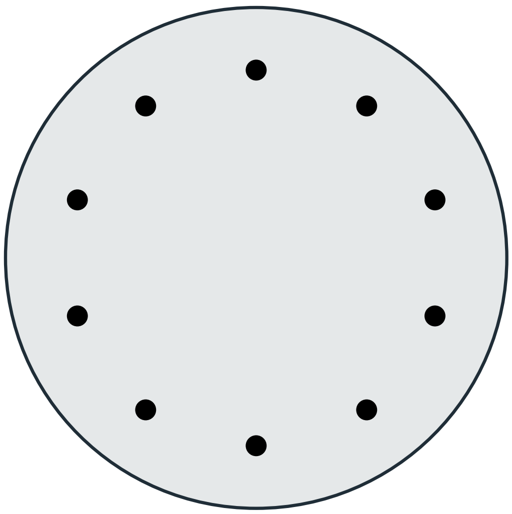
3.1 Section Figure



Diameter 24.00 in

Figure 1: Reinforced concrete section

Section summary	Value
Shape	Circular
Dimensions	Diameter 24.00 in
Clear cover	2.000 in
Bar centerline cover	3.000 in
Longitudinal bars	10 - #8
Steel area, Ast	7.900 in2
Reinforcement ratio	1.748%
Tie size	#4

4. Reinforcement - Bars Provided

Bar	X (in)	Y (in)	Area (in2)
B1	0.000	9.000	0.790
B2	5.290	7.281	0.790
B3	8.560	2.781	0.790
B4	8.560	-2.781	0.790
B5	5.290	-7.281	0.790
B6	0.000	-9.000	0.790
B7	-5.290	-7.281	0.790
B8	-8.560	-2.781	0.790
B9	-8.560	2.781	0.790
B10	-5.290	7.281	0.790

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5. Factored Loads and Corresponding Capacity Ratios

DCR is the demand moment radius divided by radial capacity at the same Pu.

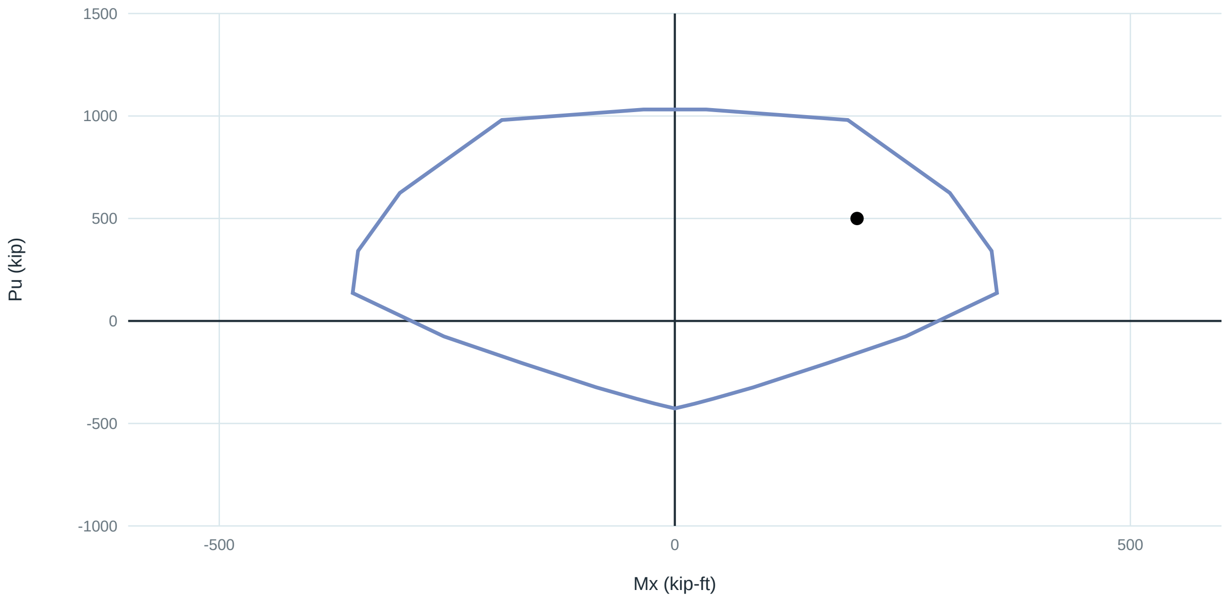
Load	Pu	Mux	Muy	Mr,cap	DCR	Status	Note
LC-C	500.00	200.00	100.00	315.63	0.708	OK	

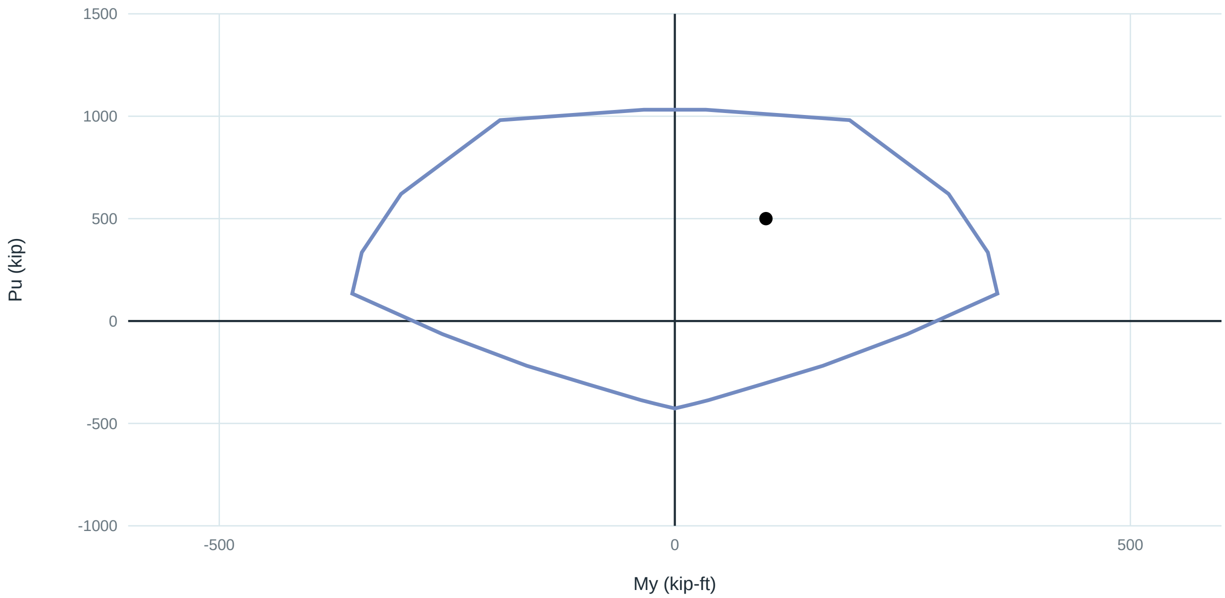
Moments are kip-ft; axial forces are kip. Compression is positive.

Beam ID: C-1 | Units: kip, kip-ft, in | Analysis: shape

6. Diagrams

6.1 Factored P-M interaction diagrams

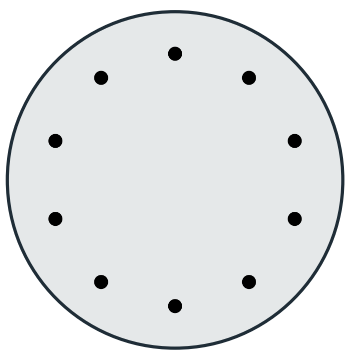




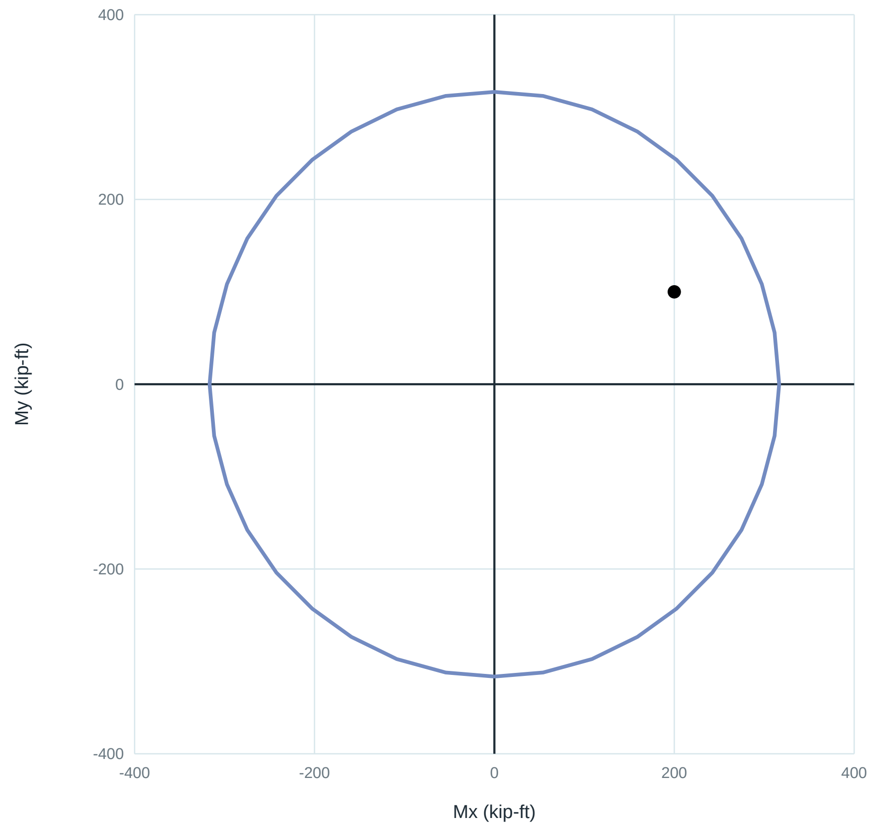
Beam ID: C-1 | Units: kip, kip-ft, in | Analysis: shape

6.2 Constant-Pu biaxial interaction

Mx-My at Pu = 500.000 kip - LC-C



Diameter 24.00 in



Load	Pu	Mux	Muy	Capacity radius	DCR	Status
LC-C	500.000	200.000	100.000	315.630	0.708	OK

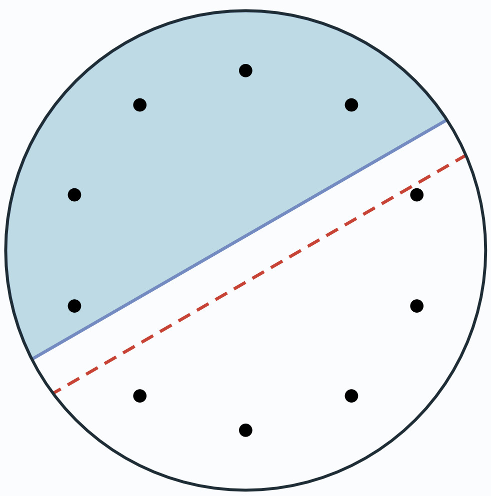
Beam ID: C-1 | Units: kip, kip-ft, in | Analysis: shape

6.3 Controlling Section Strain and Stress

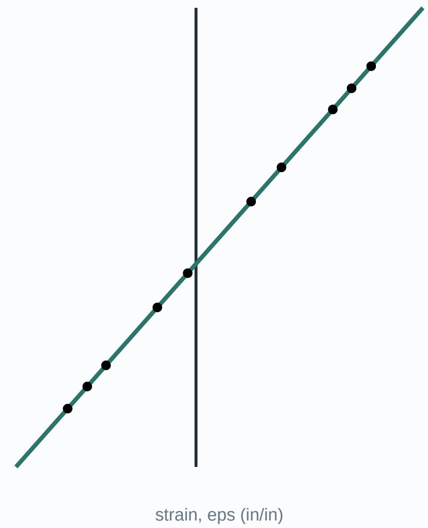
Selected load: LC-C - compression-controlled

Closest solved orientation from the 10-degree contour; moment-direction difference = 3.303 degrees.

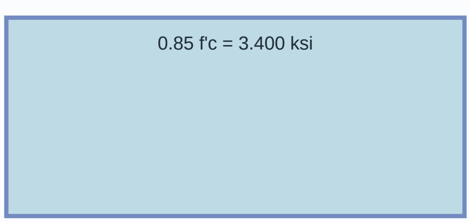
Section strain plane



Section strain distribution



ACI equivalent stress block



a = beta1 c = 11.373 in

Design idealization - not a constitutive stress-strain curve

Steel stress-strain